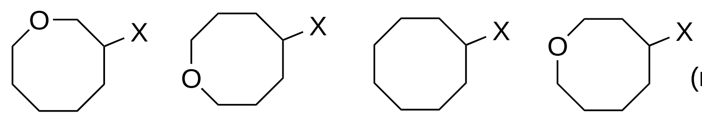
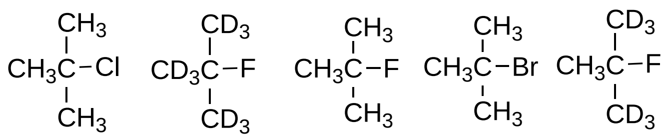
ACML100 Tutorial 4
Substitution and Elimination reactions

Q1. Arrange the following compounds in increasing order of their reactivity.

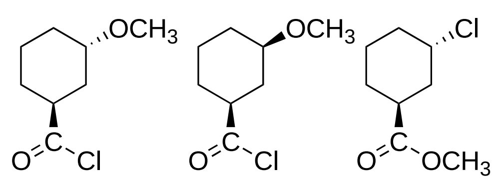
(i)  (rate of solvolysis)

I II III IV

(ii)  (E2)

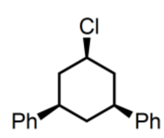
I II III IV V

Q2. Which one of the compounds (**I** or **II**) reacts with Cl^- to give (**III**)? Give the mechanism of its formation.



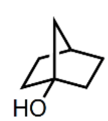
I II III

Q3. a) Explain why the compound A does not undergo an E2 elimination with strong base.



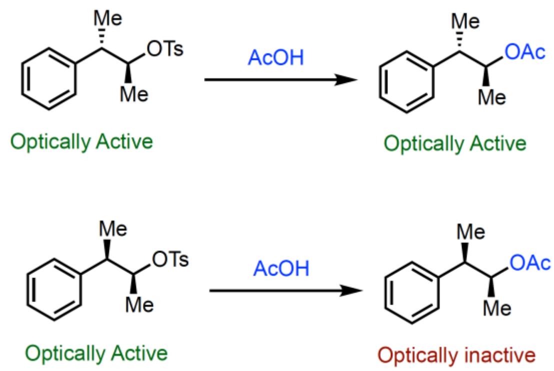
A

b) Explain why the compound B do not undergo an E1 elimination with strong acid.



B

Q4. Explain the following observation with the suitable mechanism.



Q5. Arrange the order of reactivity of the following compounds to the action of aq. alkali and suggest reasons: chlorobenzene, vinyl chloride, allyl chloride.

Q6. Arrange the following series of ions in increasing order of stability:

(i) $(CH_3)_3\,C^+$, $(CH_3)_2CH^+$, $CH_3CH_2^+$

(ii) $(C_6H_5)_3C^+$, $(C_6H_5)_2CH^+$, $C_6H_5CH_2^+$

(iii) p-$CH_3OC_6H_4CH_2^+$, p-$ClC_6H_4CH_2^+$, p-$O_2NC_6H_4CH_2^+$